

Convex Analysis and Monotone Operator Theory in Hilbert Spaces

Chapter 3: Convex Sets

Based on Bauschke & Combettes

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Chapter Overview

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Notation and Background You Need I

Throughout, $\mathcal{H}$ is a real Hilbert space (think: a vector space with an inner product $\langle \cdot | \cdot \rangle$, a norm $\|x\| = \sqrt{\langle x | x \rangle}$, and enough completeness for limits of Cauchy sequences to exist). **Whenever possible we will instantiate ideas in $\mathbb{R}^2$ with the ordinary dot product**, so you always have a concrete picture available.

- **Segment** $[x, y] = \{\alpha x + (1 - \alpha)y : \alpha \in [0, 1]\}$: every point “between” x and y on the straight line joining them. The *open* segment is $]x, y[= \{\alpha x + (1 - \alpha)y : \alpha \in]0, 1[\}$ (endpoints excluded).
- **Convex set** C : a set where $[x, y] \subset C$ whenever $x, y \in C$ – in plain words, *the straight line between any two points of C never leaves C* .
- **Convex combination**: a point $\sum_{i \in I} \alpha_i x_i$ where I is finite, each $x_i \in C$, each $\alpha_i \in [0, 1]$, and $\sum_i \alpha_i = 1$ – a weighted average of finitely many points.
- **Convex hull** $\text{conv } C$: the set of *all* convex combinations of points of C ; equivalently, the smallest convex set containing C (Proposition 3.4).

Notation and Background You Need II

- **Closure** $\overline{C}$ / **interior** $\text{int } C$: the usual topological closure and interior with respect to the norm (“strong”) topology.
- **Ball** $B(x; \rho) = \{y \in \mathcal{H} : \|y - x\| \leq \rho\}$: closed ball of center x and radius ρ .
- **Distance to a set**: $d_C(x) = \inf_{y \in C} \|x - y\|$, the distance from x to the (nonempty) set C .
- **Weak convergence** $x_n \rightharpoonup x$: for every $u \in \mathcal{H}$, $\langle x_n | u \rangle \rightarrow \langle x | u \rangle$. In $\mathbb{R}^n$ this is the same as ordinary convergence; in infinite dimensions it is strictly weaker. *Weakly closed* / *weakly compact* are the corresponding topological notions for the weak topology. A key theme of this chapter: for *convex* sets, weak and strong closedness coincide (Theorem 3.34).
- **Affine operator** $T: \mathcal{H} \rightarrow \mathcal{K}$: $T = L + b$ with L linear and b a fixed vector – i.e., linear up to a shift.

Notation and Background You Need III

- **Best approximation / projection:** $p \in C$ is a *best approximation* to x from C if $\|x - p\| = d_C(x)$ – the point of C closest to x . If every x has exactly one such p , C is a *Chebyshev set* and we write $p = P_C x$ (P_C is the *projector* onto C).
- **Bounded linear operator** $T \in \mathcal{B}(\mathcal{H}, \mathcal{K})$, its **adjoint** T^* (satisfying $\langle Tx \mid y \rangle = \langle x \mid T^*y \rangle$), and its **kernel/range** $\ker T$, $\operatorname{ran} T$: standard linear-algebra objects, now allowed to be infinite-dimensional.
- **Separation:** two sets C, D are *separated* if some nonzero $u \in \mathcal{H}$ and a hyperplane $H = \{z : \langle z \mid u \rangle = \eta\}$ have C on one side ($\sup \langle C \mid u \rangle \leq \eta$) and D on the other ($\eta \leq \inf \langle D \mid u \rangle$); *strongly separated* if the inequality chain has a positive gap.

Why convexity? The intuitive picture

Informal definition

A set $C \subset \mathcal{H}$ is **convex** if, for any two points you pick in C , the entire straight segment joining them also lies in C . No “dents,” no “holes,” no disconnected pieces that force a detour.

Convex sets are the geometric backbone of this book: projections, separation, and (later) convex functions all rest on this one simple segment condition. Figure 1 (next slide) shows the contrast: disks and triangles are convex; a crescent and a union of two disks are not, because you can find two points whose connecting segment exits the set.

Convex vs. non-convex sets

Convexity: for every $x, y \in C$, the segment $[x, y]$ stays inside C

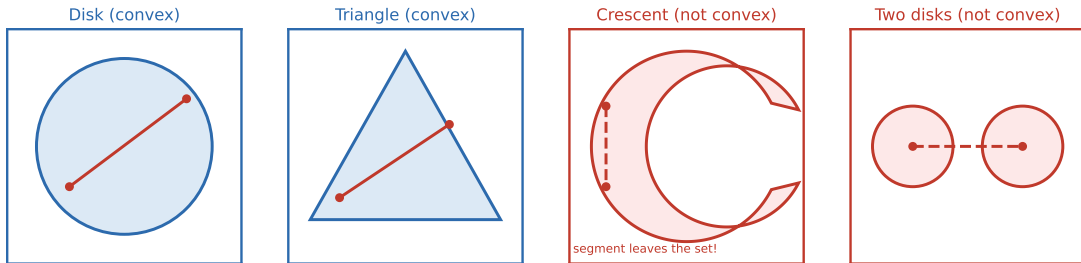


Figure: *

Left two: convex (disk, triangle). Right two: not convex (crescent, union of two disks) – the dashed red segment leaves the shaded region.

Convex combinations and the convex hull

Proposition 3.4

Let C be a subset of $\mathcal{H}$ and let D be the set of all convex combinations of points in C , i.e.,

$$D = \left\{ \sum_{i \in I} \alpha_i x_i \mid I \text{ finite, } (x_i)_{i \in I} \in C^I, (\alpha_i)_{i \in I} \in [0, 1]^I, \sum_{i \in I} \alpha_i = 1 \right\}.$$

Then $D = \text{conv } C$.

Symbols: I is a finite index set, $(x_i)_{i \in I}$ a finite family of points of C , $(\alpha_i)_{i \in I}$ nonnegative weights summing to 1. So D collects every finite weighted average of points of C ; the proposition says this coincides exactly with the convex hull of C (the smallest convex set containing C). Proof left as Exercise 3.1 – straightforward once convexity of D and minimality are checked directly from the segment definition.

Warm-up fact: intersections and images of convex sets

Before the book's propositions, here is the most basic stability property of convexity – used implicitly throughout the chapter.

Fact: an intersection of convex sets is convex

If $(C_i)_{i \in I}$ is any family of convex subsets of $\mathcal{H}$, then $\bigcap_{i \in I} C_i$ is convex.

Step-by-step proof.

- 1 Let $x, y \in \bigcap_{i \in I} C_i$ and $\alpha \in [0, 1]$. We must show $\alpha x + (1 - \alpha)y \in \bigcap_{i \in I} C_i$.
- 2 Fix an arbitrary index $i \in I$. Since $x, y \in \bigcap_{i \in I} C_i$, in particular $x \in C_i$ and $y \in C_i$.
- 3 Because C_i is convex, $\alpha x + (1 - \alpha)y \in C_i$.
- 4 Since $i \in I$ was arbitrary, $\alpha x + (1 - \alpha)y \in C_i$ for every $i \in I$, i.e., $\alpha x + (1 - \alpha)y \in \bigcap_{i \in I} C_i$.

This is exactly why, e.g., an affine subspace (an intersection of hyperplanes) or a polyhedron (an intersection of half-spaces) is convex.

Affine images and preimages preserve convexity

Proposition 3.5

Let $\mathcal{K}$ be a real Hilbert space, let $T: \mathcal{H} \rightarrow \mathcal{K}$ be an affine operator, and let $C \subset \mathcal{H}$, $D \subset \mathcal{K}$ be convex. Then $T(C)$ and $T^{-1}(D)$ are convex (subsets of $\mathcal{K}$ and $\mathcal{H}$, respectively).

Symbols: T affine means $T = L + b$ for a linear map L and fixed vector b ; $T(C) = \{Tx : x \in C\}$ is the image, $T^{-1}(D) = \{x \in \mathcal{H} : Tx \in D\}$ the preimage.

Why it is true (sketch): an affine map sends segments to segments, $T([x, y]) = [Tx, Ty]$. So if $Tx, Ty \in T(C)$ with $x, y \in C$, convexity of C gives $[x, y] \subset C$, hence $[Tx, Ty] = T([x, y]) \subset T(C)$: the image is convex. Symmetrically, if $x, y \in T^{-1}(D)$ then $Tx, Ty \in D$, so $[Tx, Ty] \subset D$ by convexity of D , and $T([x, y]) = [Tx, Ty] \subset D$ shows $[x, y] \subset T^{-1}(D)$.

Proposition 3.6

Let $(C_i)_{i \in I}$ be a totally ordered finite family of m convex subsets of $\mathcal{H}$. Then:

- (i) $\times_{i \in I} C_i$ is convex (as a subset of $\mathcal{H}^m$).
- (ii) For every $(\alpha_i)_{i \in I} \in \mathbb{R}^m$, $\sum_{i \in I} \alpha_i C_i$ is convex.

Symbols: $\times_{i \in I} C_i$ is the Cartesian product $C_1 \times \cdots \times C_m$; $\sum_i \alpha_i C_i = \{\sum_i \alpha_i x_i : x_i \in C_i\}$ is a Minkowski-type weighted sum.

Proof idea. (i) is a direct segment check coordinate-by-coordinate. (ii) follows from (i) and Proposition 3.5: the map $L: \mathcal{H}^m \rightarrow \mathcal{H}, (x_i)_{i \in I} \mapsto \sum_{i \in I} \alpha_i x_i$ is linear (hence affine), and $\sum_{i \in I} \alpha_i C_i = L(\times_{i \in I} C_i)$ is the image of a convex set under an affine map – convex by Proposition 3.5.

A rigidity fact about constant-norm convex sets

Proposition 3.7

Let C be a nonempty convex subset of $\mathcal{H}$ and $\beta \in \mathbb{R}_+$. Suppose $\|x\| = \beta$ for every $x \in C$. Then C is a singleton.

Idea: if $\beta = 0$ then $C = \{0\}$ trivially. If $\beta > 0$ and C had two distinct points x, y , the midpoint $z = (x + y)/2$ would lie in C by convexity, so $\|z\| = \beta$. But the (strict) triangle-type inequality for norms coming from an inner product (Corollary 2.16) forces $\|z\| < \|x\| = \beta$ whenever $x \neq y$ – a contradiction. So C cannot contain two distinct points: a sphere contains no nontrivial convex subset except single points.

Projections: intuition first

Given a point x and a target set C , we ask: *what point of C is closest to x ?* That closest point is called a **projection** of x onto C . Sometimes there is none, sometimes several, and (as we will prove) exactly one whenever C is nonempty, closed, and *convex*. This uniqueness is the single most important structural fact about convex sets in a Hilbert space, and everything downstream (Chapters 4, 20–30) is built on it.

Definition 3.8

Let C be a nonempty subset of $\mathcal{H}$, $x \in \mathcal{H}$, $p \in C$. Then p is a *best approximation* to x from C (a *projection* of x onto C) if $\|x - p\| = d_C(x)$. If every point of $\mathcal{H}$ has at least one projection onto C , C is *proximal*; if every point has *exactly* one, C is a *Chebyshev set*, and the map sending x to its unique projection is the *projector* P_C .

Example: low-rank matrix approximation (Eckart–Young)

Example 3.9 (Eckart–Young theorem)

Let $\mathcal{H}$ be the space of $M \times N$ matrices with the Frobenius inner product, $m = \min\{M, N\} \geq 2$, $q \in \{1, \dots, m-1\}$, and $C = \{A \in \mathcal{H} : \text{rank } A \leq q\}$. Given A with SVD $A = U\Sigma V^\top$ and singular values $\sigma_1(A) \geq \dots \geq \sigma_m(A) \geq 0$, the truncated matrix $P = U\Sigma_q V^\top$, with $\Sigma_q = \text{Diag}(\sigma_1(A), \dots, \sigma_q(A), 0, \dots, 0)$, is a projection of A onto C ; it is the unique one iff $\sigma_{q+1}(A) \neq \sigma_q(A)$.

Reading it: C is *not* convex in general (it's a union of rank strata), so C is proximal but not Chebyshev – exactly the distinction Definition 3.8 draws. The proof expands $\|A - B\|_F^2$ in the rotated coordinates $X = U^\top B V$ and minimizes term by term, showing the best rank- q truncation simply zeroes out the smallest singular values.

Example: projection onto a finite orthonormal span

Example 3.10

Let $\{e_i\}_{i \in I}$ be a finite orthonormal set, $V = \text{span}\{e_i\}_{i \in I}$, $x \in \mathcal{H}$. Then V is a Chebyshev set with

$$P_C = P_V : \quad P_V x = \sum_{i \in I} \langle x | e_i \rangle e_i, \quad d_V(x) = \sqrt{\|x\|^2 - \sum_{i \in I} |\langle x | e_i \rangle|^2}.$$

Symbols: $\langle x | e_i \rangle$ is the coordinate of x along the orthonormal direction e_i ; the formula says the closest point in the span is obtained by keeping only those coordinates, and the leftover distance is the norm of what got discarded. **Proof sketch (eq. 3.8):** expand

$\|x - \sum_i \alpha_i e_i\|^2 = \|x\|^2 - \sum_i |\langle x | e_i \rangle|^2 + \sum_i |\alpha_i - \langle x | e_i \rangle|^2$ using orthonormality; this is minimized exactly when $\alpha_i = \langle x | e_i \rangle$ for every i .

Closedness is necessary, but not sufficient, for proximality

Remark 3.11

- (i) $\overline{C} = \{x \in \mathcal{H} : d_C(x) = 0\}$, so no point of $\overline{C} \setminus C$ has a projection onto C : *every proximal set (in particular every Chebyshev set) must be closed.*
- (ii) Every finite-dimensional linear subspace is a Chebyshev set (it has a finite orthonormal basis by Gram–Schmidt, so Example 3.10 applies), hence closed by (i).

Example 3.13 (closed but not proximal)

Let $(e_n)_{n \in \mathbb{N}}$ be orthonormal in an infinite-dimensional $\mathcal{H}$, $(\alpha_n) \downarrow 1$ in $]1, +\infty[$, and $C = \{\alpha_n e_n\}_{n \in \mathbb{N}}$. Then C is closed (every convergent sequence in C is eventually constant), yet 0 has *no* projection onto C , since $d_C(0) = 1 < \alpha_n = \|0 - \alpha_n e_n\|$ for every n : the infimum distance 1 is approached but never attained.

Sufficient conditions for proximality

Proposition 3.12

If $\mathcal{H}$ is finite-dimensional and C is a Chebyshev subset of $\mathcal{H}$, then P_C is continuous.

Proposition 3.14

Every nonempty *weakly closed* subset of $\mathcal{H}$ is proximal.

Why weak closedness helps: minimizing $\|x - \cdot\|$ over C reduces to minimizing over $D = C \cap B(x; \|x - z\|)$ for any fixed $z \in C$; D is weakly compact (closed ball is weakly compact, Fact 2.34, intersected with a weakly closed set), and the norm is weakly lower semicontinuous, so a minimizer exists by the Weierstrass-type Theorem 1.29 in the weak topology.

Corollary 3.15

If $\mathcal{H}$ is finite-dimensional, C is proximal $\iff C$ is closed (weak and strong closedness coincide in finite dimensions).

The Projection Theorem: the central result of this section

Theorem 3.16 (Projection theorem)

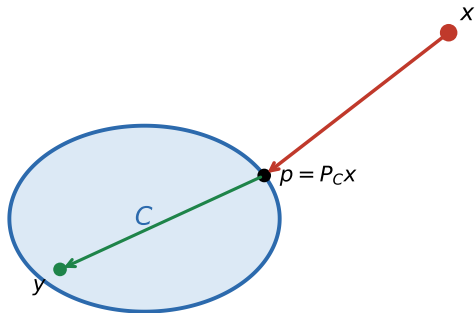
Let C be a nonempty closed *convex* subset of $\mathcal{H}$. Then C is a Chebyshev set, and for every $x, p \in \mathcal{H}$,

$$p = P_C x \iff \left[p \in C \text{ and } (\forall y \in C) \langle y - p \mid x - p \rangle \leq 0 \right].$$

Symbols: the right-hand condition says p is a projection exactly when, from p , every other point y of C makes a right angle or an *obtuse* angle with the direction toward x – see the figure on the next slide. This is remarkable: existence *and* uniqueness of the closest point, for every x , follow purely from C being closed and convex (Chebyshev sets need not be convex in general, cf. Remark 3.17, but every nonempty closed convex set automatically is one).

Picture: the obtuse-angle characterization

$\langle y - p, x - p \rangle \leq 0$ for every $y \in C$
(the angle at p is right or obtuse)



Step-by-step proof of the Projection Theorem (existence & uniqueness)

Setup. Fix $x \in \mathcal{H}$; we must produce a unique closest point.

① By definition of $d_C(x) = \inf_{y \in C} \|x - y\|$, choose a sequence (y_n) in C with $\|y_n - x\| \rightarrow d_C(x)$ (a *minimizing sequence*).

② **Cauchy via Apollonius' identity.** For $m, n \in \mathbb{N}$, since C is convex, $(y_n + y_m)/2 \in C$, so $\|x - (y_n + y_m)/2\| \geq d_C(x)$. Apollonius' identity (Lemma 2.12(iv)) gives

$$\|y_n - y_m\|^2 = 2\|y_n - x\|^2 + 2\|y_m - x\|^2 - 4\|x - \frac{y_n + y_m}{2}\|^2 \leq 2\|y_n - x\|^2 + 2\|y_m - x\|^2 - 4d_C(x)^2.$$

As $m, n \rightarrow \infty$ the right side $\rightarrow 0$, so (y_n) is Cauchy.

③ **Existence.** C is closed and $\mathcal{H}$ complete, so C is complete; the Cauchy sequence (y_n) converges to some $p \in C$. Continuity of $\|\cdot - x\|$ gives $\|p - x\| = \lim \|y_n - x\| = d_C(x)$: p is a projection.

④ **Uniqueness.** If $q \in C$ also satisfies $\|q - x\| = d_C(x)$, the same Apollonius computation with $y_n \equiv p$, $y_m \equiv q$ gives $\|p - q\|^2 \leq 4d_C(x)^2 - 4\|x - (p + q)/2\|^2 \leq 0$ (since $(p + q)/2 \in C$ forces $\|x - (p + q)/2\| \geq d_C(x)$), so $p = q$.

⑤ **The characterization.** For $y \in C$, $\alpha \in [0, 1]$, set $y_\alpha = \alpha y + (1 - \alpha)p \in C$ (convexity). Then

$$\begin{aligned} \|x - p\| = d_C(x) &\iff (\forall y \in C)(\forall \alpha \in [0, 1]) \|x - p\| \leq \|x - p - \alpha(y - p)\| \\ &\iff (\forall y \in C)(\forall \alpha \in [0, 1]) 0 \leq -2\alpha \langle y - p | x - p \rangle + \alpha^2 \|y - p\|^2. \end{aligned}$$

Dividing by $\alpha > 0$ and letting $\alpha \downarrow 0$ forces $\langle y - p | x - p \rangle \leq 0$; conversely this inequality (for all y) plainly implies the displayed chain with $\alpha = 1$. This proves the equivalence

Numeric example: projecting onto a disk

Running example. Take $C = B(0; 2) \subset \mathbb{R}^2$ (closed disk of radius $\rho = 2$).

Example 3.18

For $\rho \in \mathbb{R}_{++}$ and $C = B(0; \rho)$: $P_C x = \frac{\rho}{\max\{\|x\|, \rho\}} x$ and $d_C(x) = \max\{\|x\| - \rho, 0\}$.

Concrete numbers. Let $x = (3, 4)$, so $\|x\| = 5 > \rho = 2$. Then

$$P_C x = \frac{2}{5}(3, 4) = (1.2, 1.6), \quad d_C(x) = 5 - 2 = 3.$$

Check the obtuse-angle rule: take any $y \in C$, e.g. $y = (-2, 0)$. Then $x - p = (1.8, 2.4)$ and $y - p = (-3.2, -1.6)$, and $\langle y - p \mid x - p \rangle = (-3.2)(1.8) + (-1.6)(2.4) = -5.76 - 3.84 = -9.6 \leq 0$. If instead $x = (1, 1)$ (with $\|x\| = \sqrt{2} \leq \rho$), then $P_C x = x$ itself and $d_C(x) = 0$: points already inside a closed convex set are their own projection.

Python: verifying the projection formula numerically

```
1 import numpy as np
2
3 def project_disk(x, rho=2.0):          # Example 3.18:  $P_C$  for  $C = B(0; \rho)$ 
4     n = np.linalg.norm(x)
5     return x * (rho / max(n, rho))
6
7 def dist_disk(x, rho=2.0):
8     return max(np.linalg.norm(x) - rho, 0.0)
9
10 x = np.array([3.0, 4.0])
11 p = project_disk(x)
12 print("P_C(x) =", p, "   d_C(x) =", dist_disk(x))
13
14 # obtuse-angle check (Theorem 3.16) against several sample points y in C
15 rng = np.random.default_rng(0)
16 for _ in range(5):
17     y = rng.normal(size=2); y = y / max(np.linalg.norm(y), 1) * 2.0 # y in C
18     lhs = np.dot(y - p, x - p)
19     print(f"y={np.round(y,2)}   <y-p, x-p> = {lhs:.4f}   (should be <= 0)")
```

Output (excerpt): $P_C(x) = [1.2 \ 1.6]$ $d_C(x) = 3.0$, and every sampled $\langle y-p, x-p \rangle$ value is ≤ 0 , confirming Theorem 3.16 numerically for this running example.

Elementary consequences of the Projection Theorem

Proposition 3.19 (translation)

If C is nonempty closed convex and $x, y \in \mathcal{H}$, then $P_{y+C} x = y + P_C(x - y)$: projecting onto a shifted set is the same as shifting, projecting onto the original set, and shifting back.

Proposition 3.20 (nested sets)

If $(C_n)_{n \in \mathbb{N}}$ are nonempty bounded closed convex sets with $C_{n+1} \subset C_n$ for every n , then $\bigcap_{n \in \mathbb{N}} C_n \neq \emptyset$. (Boundedness matters: $([n, +\infty[)_{n \in \mathbb{N}}$ in $\mathcal{H} = \mathbb{R}$ is a decreasing chain of closed convex sets with empty intersection.)

Proposition 3.21 (idempotence along the ray to x)

For $\lambda \in \mathbb{R}_+$, $P_C(P_C x + \lambda(x - P_C x)) = P_C x$: moving from $P_C x$ further toward (or away from, on the far side of) x does not change the projection.

Projectors onto affine subspaces and hyperplanes

Corollary 3.22

Let C be a *closed affine subspace*. Then (i) $p = P_C x$ iff $p \in C$ and $\langle y - z \mid x - p \rangle = 0$ for all $y, z \in C$ (equality, not just $\leq$ – because an affine subspace has no “far side”); (ii) P_C is an *affine operator*.

Example 3.23 (hyperplane)

For $u \in \mathcal{H} \setminus \{0\}$, $\eta \in \mathbb{R}$, $C = \{x : \langle x \mid u \rangle = \eta\}$:

$$P_C x = x + \frac{\eta - \langle x \mid u \rangle}{\|u\|^2} u, \quad d_C(x) = \frac{|\langle x \mid u \rangle - \eta|}{\|u\|}.$$

Numeric instance (extends the running example). Let $u = (1, 1)$, $\eta = 3$, so $C = \{(\xi_1, \xi_2) : \xi_1 + \xi_2 = 3\}$. For $x = (0, 0)$: $\langle x \mid u \rangle = 0$, $\|u\|^2 = 2$, so

$$P_C x = (0, 0) + \frac{3 - 0}{2}(1, 1) = (1.5, 1.5), \quad d_C(x) = \frac{|0 - 3|}{\sqrt{2}} = \frac{3}{\sqrt{2}} \approx 2.121.$$

Sanity check: $1.5 + 1.5 = 3$, so $(1.5, 1.5) \in C$ as required, and $\|(0, 0) - (1.5, 1.5)\| = \sqrt{4.5} \approx 2.121$. ✓

Projectors onto linear subspaces: the orthogonal split

Corollary 3.24

Let V be a closed linear subspace, $x \in \mathcal{H}$. Then, among other properties: (i) $p = P_V x \iff (p, x - p) \in V \times V^\perp$; (iv) $V^{\perp\perp} = V$; (v) $P_{V^\perp} = \text{Id} - P_V$; (vii) $\|x\|^2 = \|P_V x\|^2 + \|P_{V^\perp} x\|^2 = d_V(x)^2 + d_{V^\perp}(x)^2$.

Reading it: every x splits uniquely as $x = P_V x + P_{V^\perp} x$, a component *in* V plus a component perpendicular to V – the Hilbert-space Pythagorean theorem in (vii) is exactly $\|x\|^2 = (\text{in-}V \text{ part})^2 + (\text{perp part})^2$.

Proposition 3.25

For any nonempty $C \subset \mathcal{H}$ with $V = \overline{\text{span}} C$, the set-valued projector satisfies $\Pi_C = \Pi_C \circ P_V$: projecting onto C can always be done by first projecting onto the (smaller, linear) closed span of C .

Application: least-squares solutions and the generalized inverse

Definition 3.26 / Proposition 3.27

Given $T \in \mathcal{B}(\mathcal{H}, \mathcal{K})$ with $\text{ran } T$ closed and $y \in \mathcal{K}$, x is a *least-squares solution* of $Tz = y$ if $\|Tx - y\| = \min_{z \in \mathcal{H}} \|Tz - y\|$. Equivalently: $Tx = P_{\text{ran } T}y$, or (the *normal equation*) $T^*Tx = T^*y$.

Why: minimizing $\|Tz - y\|$ over all z is the same as minimizing $\|r - y\|$ over $r \in \text{ran } T$ – exactly a projection problem onto the closed linear subspace $\text{ran } T$, so Theorem 3.16 applies via Corollary 3.24(i): $r - y \perp \text{ran } T \iff T^*(Tx - y) = 0$.

Definition 3.28 (Moore–Penrose inverse)

The set of least-squares solutions is a nonempty closed affine subspace; its unique minimal-norm element is denoted $T^\dagger y$, defining the operator $T^\dagger: \mathcal{K} \rightarrow \mathcal{H}$.

Example 3.29: if T^*T is invertible, $T^\dagger = (T^*T)^{-1}T^*$ – the familiar least-squares formula from linear regression.

Properties of the generalized inverse

Proposition 3.30 / Theorem 3.31 (Moore–Desoer–Whalen) / Corollary 3.32

For $T \in \mathcal{B}(\mathcal{H}, \mathcal{K})$ with $\text{ran } T$ closed: $T^\dagger \in \mathcal{B}(\mathcal{K}, \mathcal{H})$, $\text{ran } T^\dagger = \text{ran } T^*$, $P_{\text{ran } T} = TT^\dagger$, $P_{\ker T} = \text{Id} - T^\dagger T$, and (Penrose relations) $T^\dagger T^\dagger = T$, $T^\dagger T^* = T^* T^\dagger$, $T^* T^* T^\dagger = T^\dagger T$.

Big picture: $T^\dagger$ generalizes matrix inversion to possibly-non-invertible (even non-square) bounded operators with closed range, in exactly the way "solve $Ax = b$ in the least-squares, minimum-norm sense" generalizes "solve $Ax = b$ " – the workhorse behind pseudoinverses in numerical linear algebra.

Why weak vs. strong closedness matters

In infinite dimensions, weak and strong (norm) closedness can differ for *general* sets.

Example 3.33

With (e_n) orthonormal, $\alpha_n \uparrow +\infty$, $C = \{\alpha_n e_n\}_{n \in \mathbb{N}}$ is closed and weakly sequentially closed, but *not* weakly closed: 0 lies in the weak closure of C even though $0 \notin C$ and no sequence in C converges weakly to 0 (weak *sequential* closedness is about sequences; weak closedness is about the finer weak-neighborhood topology, and nets, not just sequences).

The next theorem shows this pathology is impossible once C is convex – a major reason convexity is so useful.

Theorem 3.34: all closedness notions coincide for convex sets

Theorem 3.34

Let C be a convex subset of $\mathcal{H}$. Then the following are equivalent: (i) C is weakly sequentially closed; (ii) C is sequentially closed; (iii) C is closed; (iv) C is weakly closed.

Proof sketch of the interesting direction (iii) $\Rightarrow$ (iv). Let a net $(x_a)_{a \in A}$ in C converge weakly to some $x \in \mathcal{H}$. Since C is closed and convex, Theorem 3.16 gives $\langle x_a - P_C x \mid x - P_C x \rangle \leq 0$ for every a . Passing to the (weak) limit in a termwise, $\langle x - P_C x \mid x - P_C x \rangle = \|x - P_C x\|^2 \leq 0$, forcing $x = P_C x \in C$. So a weak limit of points of C must already lie in C : C is weakly closed. The remaining implications are general topology / metric-space facts (equivalences (ii) $\Leftrightarrow$ (iii) always hold in a metric space; (i) $\Rightarrow$ (ii) and (iv) $\Rightarrow$ (i) use general weak-topology facts).

Consequences: Mazur's lemma and weak compactness

Corollary 3.35 / Corollary 3.36 (Mazur's lemma)

If C is closed and convex and $x_n \rightharpoonup x$ with $x_n \in C$, then $x \in C$. More generally, if $x_n \rightharpoonup x$ (no convexity assumed on the sequence's set), there exist *convex combinations* $y_n = \sum_{k=0}^n \alpha_{k,n} x_k \rightarrow x$ *strongly*.

Idea: apply Theorem 3.34 to $\overline{\text{conv}} C$ (closed, convex, hence weakly closed by the theorem), so the weak limit x lies in $\overline{\text{conv}}\{x_n\}$, i.e., is a strong limit of actual convex combinations of the x_n 's.

Theorem 3.37 / Corollary 3.38

A bounded closed convex C is weakly compact and weakly sequentially compact: every sequence in C has a weakly convergent subsequence, with weak limit in C .

Proposition 3.39

If $C_1, \dots, C_m$ are compact (resp. weakly compact), then $\text{conv} \bigcup_i C_i$ is compact (resp. weakly compact).

Sums of convex sets need not be closed

Example 3.40 (finite dimensions!)

In $\mathbb{R}^2$, $C = \mathbb{R} \times \{0\}$ and $D = \{(\xi_1, \xi_2) \in \mathbb{R}_{++}^2 : \xi_1 \xi_2 \geq 1\}$ are both closed and convex, yet $C + D = \mathbb{R} \times \mathbb{R}_{++}$ is *open* – not closed.

Proposition 3.42 (a fix: boundedness of one summand)

If C, D are nonempty closed convex and D is *bounded*, then $C + D$ is nonempty, closed, and convex.

Why boundedness saves us: take $x_n + y_n \rightarrow z$ with $x_n \in C$, $y_n \in D$. By Corollary 3.38, a subsequence $y_{k_n} \rightarrow y \in D$ (boundedness of D gives weak compactness); then $x_{k_n} = z - y_{k_n} + (\text{vanishing}) \rightarrow z - y$, and Corollary 3.35 places $z - y \in C$ (closed convex $\Rightarrow$ weakly closed). Hence $z = (z - y) + y \in C + D$.

Segments from the interior reach into the interior

Proposition 3.44

Let C be convex. Then for every $x \in \text{int } C$ and $y \in \overline{C}$, $[x, y[\subset \text{int } C$.

Step-by-step proof.

- 1 If $x = y$ the claim is trivial, so assume $x \neq y$ and fix $z = \alpha x + (1 - \alpha)y \in [x, y[$ with $\alpha \in]0, 1[$.
- 2 Since $x \in \text{int } C$, choose $\varepsilon > 0$ with $B(x; \varepsilon(2 - \alpha)/\alpha) \subset C$.
- 3 Since $y \in \overline{C}$, $y \in C + B(0; \varepsilon)$ (points of the closure are approximated within any radius by points of C).
- 4 Using convexity to expand $B(z; \varepsilon)$ as an affine combination of the two previous balls:

$$B(z; \varepsilon) = \alpha x + (1 - \alpha)y + B(0; \varepsilon) \subset \alpha B(x; \varepsilon(2 - \alpha)/\alpha) + (1 - \alpha)C \subset \alpha C + (1 - \alpha)C = C,$$

using Exercise 3.3's fact ($\alpha C + (1 - \alpha)C = C$ for convex C).

- 5 So a whole ball around z lies in C : $z \in \text{int } C$. Since $z \in [x, y[$ was arbitrary, $[x, y[\subset \text{int } C$.

Closure and interior of a convex set are convex

Proposition 3.45

Let C be convex. Then (i) $\overline{C}$ is convex; (ii) $\text{int } C$ is convex; (iii) if $\text{int } C \neq \emptyset$, then $\text{int } C = \text{int } \overline{C}$ and $\overline{C} = \overline{\text{int } C}$.

Why: (i) is a direct sequential-limit argument (limits of convex combinations of converging sequences are convex combinations of the limits). (ii) follows immediately from Proposition 3.44: if $x, y \in \text{int } C \subset \overline{C}$, the whole segment $[x, y[$ — and by symmetry all of $[x, y]$ — lies in $\text{int } C$. Part (iii) again leans on Proposition 3.44, letting a boundary-approaching point of $\overline{C}$ be reached along segments from an interior point.

Proposition 3.46

For any subset C : $\text{conv } \overline{C} \subset \overline{\text{conv } C} = \overline{\text{conv } \overline{C}}$. (In general $\text{conv } \overline{C} \neq \overline{\text{conv } C}$: Example 3.47 exhibits $C = \overline{C}$ in $\mathbb{R}^2$ with $\text{conv } C = \mathbb{R} \times \mathbb{R}_{++}$ but $\overline{\text{conv } C} = \mathbb{R} \times \mathbb{R}_+$.)

A closedness fact combining closure and interior

Proposition 3.48

Let C, D be convex with C closed and $C \cap \text{int } D \neq \emptyset$. Then $\overline{C \cap \text{int } D} = C \cap \overline{D}$.

Idea: “ $\subset$ ” is immediate from Proposition 3.45(iii) applied inside D . For “ $\supset$ ”, fix $x \in C \cap \text{int } D$ and $y \in C \cap \overline{D}$; convexity of C gives $[x, y] \subset C$, while Proposition 3.44 (applied to D) gives $[x, y[\subset \text{int } D$. Letting $\alpha \downarrow 0$ along $z_\alpha = \alpha x + (1 - \alpha)y \in C \cap \text{int } D$ shows $y = \lim_{\alpha \downarrow 0} z_\alpha \in \overline{C \cap \text{int } D}$.

Python: illustrating Mazur's lemma numerically

```
1 import numpy as np
2
3 # A sequence converging weakly (here: in finite dim, weak = strong) to 0,
4 # but built to highlight that Mazur's lemma gives STRONG convex combos.
5 rng = np.random.default_rng(1)
6 N = 6
7 xs = [np.array([np.cos(2*np.pi*k/N), np.sin(2*np.pi*k/N)]) for k in range(N)]
8 # xs sit on the unit circle (a compact, non-convex set); their average IS
9 # a convex combination and lies strictly inside conv{xs} (Mazur-style idea)
10 alphas = np.ones(N) / N
11 y = sum(a * x for a, x in zip(alphas, xs))
12 print("points on the unit circle:", [np.round(x, 3) for x in xs])
13 print("equal-weight convex combination y =", np.round(y, 6))
14 print("||y|| =", np.linalg.norm(y), " (near 0: symmetric cancellation)")
```

Output (excerpt): the six points lie on the unit circle (not convex!), yet the single convex combination $y \approx (0, 0)$ lands deep *inside* $\text{conv}\{x_k\}$ – concretely illustrating how convex combinations of a sequence can converge to a point that no individual term of the sequence approaches, the phenomenon underlying Mazur's lemma (Corollary 3.36).

Separation: intuition first

Informal idea

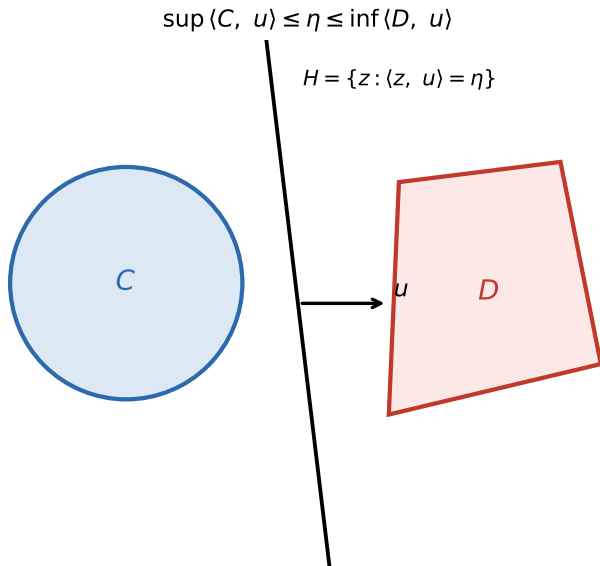
Two sets are *separated* if you can slide a hyperplane (a flat “wall” of one dimension less than the ambient space) between them so that one set lies entirely on one side and the other entirely on the other side. *Strongly separated* means there is even a gap – the wall can be thickened into a slab with neither set touching it.

Definition 3.49

$C, D \subset \mathcal{H}$ are *separated* if there exists $u \in \mathcal{H} \setminus \{0\}$ with $\sup \langle C \mid u \rangle \leq \inf \langle D \mid u \rangle$; *strongly separated* if this inequality is strict (with a uniform positive gap). A point x is (strongly) separated from D if $\{x\}$ and D are.

This is exactly the picture in Figure 3.2 (reproduced next): a line H with C pushed to one side, D to the other, u the outward normal direction.

Picture: separating two disjoint convex sets



Theorem 3.50: strong separation from a closed convex set

Theorem 3.50

Let C be nonempty closed convex and $x \in \mathcal{H} \setminus C$. Then x is *strongly* separated from C .

Step-by-step proof.

- 1 Since $x \notin C$ and C is closed convex, Theorem 3.16 gives a well-defined projection $P_C x$; set $u = x - P_C x$. Because $x \neq P_C x$ (as $x \notin C$), $u \neq 0$.
- 2 Fix any $y \in C$. The characterization (3.10) from Theorem 3.16 states $\langle y - P_C x \mid x - P_C x \rangle \leq 0$, i.e., $\langle y - P_C x \mid u \rangle \leq 0$.
- 3 Expand: $\langle y - P_C x \mid u \rangle = \langle y \mid u \rangle - \langle P_C x \mid u \rangle$, and note $\langle P_C x \mid u \rangle = \langle x - u \mid u \rangle = \langle x \mid u \rangle - \|u\|^2$. Substituting, $\langle y \mid u \rangle - \langle x \mid u \rangle + \|u\|^2 \leq 0$, i.e., $\langle y \mid u \rangle \leq \langle x \mid u \rangle - \|u\|^2$.
- 4 Since $y \in C$ was arbitrary, $\sup\{\langle y \mid u \rangle \mid y \in C\} \leq \langle x \mid u \rangle - \|u\|^2 < \langle x \mid u \rangle = \inf\{\langle x \mid y \rangle \mid y \in C\}$: a strict gap of size $\|u\|^2 > 0$ separates x from C .

This is a direct, quantitative payoff of the Projection Theorem: the separating direction u is literally “ x minus its own projection.”

Corollaries: separating two sets

Corollary 3.51

If $C, D \neq \emptyset$, $C \cap D = \emptyset$, and $C - D$ is closed and convex, then C, D are strongly separated.

Why: $C \cap D = \emptyset \iff 0 \notin C - D$. Apply Theorem 3.50 to the point 0 and the closed convex set $C - D$: strong separation of $\{0\}$ from $C - D$ translates exactly (by Definition 3.49) into strong separation of C from D .

Corollary 3.52

If C, D are nonempty closed convex, disjoint, and D is bounded, then C, D are strongly separated.
(Combine Proposition 3.42: $C - D$ is closed convex when D is bounded, then apply Corollary 3.51.)

Theorem 3.53 (finite dimensions)

If $\mathcal{H}$ is finite-dimensional and C, D are nonempty closed convex disjoint sets (boundedness of D not required), then C, D are (plainly, not necessarily strongly) separated – obtained via Corollary 3.52 applied to $D_n = D \cap B(0; n)$ and a compactness argument extracting a convergent subsequence of unit normal directions.

Infinite dimensions: separation can fail

Example 3.54

In an infinite-dimensional $\mathcal{H}$, there exist two closed affine (hence convex) subspaces U, V with $U \cap V = \emptyset$ that are *not* separated at all.

Why this matters: Theorem 3.53's separation guarantee for merely closed (not necessarily bounded) disjoint convex sets genuinely needs finite dimensions – Example 3.54 shows the boundedness or finite-dimensionality hypotheses in Corollary 3.52 / Theorem 3.53 cannot simply be dropped. The construction reuses the pathological sum-not-closed example (Example 3.41): $C + D$ dense but not closed lets one build affine sets whose difference is dense in $\mathcal{H}$, so no direction u can keep $\langle U \mid u \rangle$ and $\langle V \mid u \rangle$ from overlapping.

Chapter 3 in one slide

- **3.1 Convexity:** segments stay inside; convex hull = all finite convex combinations (Prop. 3.4); convexity survives affine images/preimages, products, and weighted sums (Props. 3.5, 3.6).
- **3.2 Best approximation:** nonempty closed *convex* sets are always Chebyshev (**Projection Theorem 3.16**), with the obtuse-angle characterization $\langle y - p \mid x - p \rangle \leq 0$; feeds directly into least-squares solutions and the Moore–Penrose inverse.
- **3.3 Topology:** for convex sets, *all* notions of closedness (closed, sequentially closed, weakly closed, weakly sequentially closed) coincide (Thm. 3.34); bounded closed convex sets are weakly compact (Thm. 3.37); segments from $\text{int } C$ into $\overline{C}$ stay in $\text{int } C$ (Prop. 3.44).
- **3.4 Separation:** a point outside a closed convex set is always *strongly* separated from it (Thm. 3.50), via $u = x - P_C x$; two disjoint closed convex sets are separated whenever one is bounded, or (in finite dimensions) unconditionally (Thm. 3.53).

Running numeric example used throughout: the triangle $T = \text{conv}\{(0, 0), (4, 0), (0, 3)\}$, the disk $B(0; 2)$, and the hyperplane $\{\xi_1 + \xi_2 = 3\}$ in $\mathbb{R}^2$.